



Sandia
National
Laboratories

Exceptional service in the national interest

ME469: Numerical Methods and Errors

Stefan P. Domino^{1,2}

¹ Computational Thermal and Fluid Mechanics, Sandia National Laboratories

² Institute for Computational and Mathematical Engineering, Stanford

This presentation has been authored by an employee of National Technology & Engineering Solutions of Sandia, LLC under Contract No. DE-NA0003525 with the U.S. Department of Energy (DOE). The employee owns all right, title and interest in and to the presentation and is solely responsible for its contents. The United States Government retains and the publisher, by accepting the article for publication, acknowledges that the United States Government retains a non-exclusive, paid-up, irrevocable, world-wide license to publish or reproduce the published form of this article or allow others to do so, for United States Government purposes. The DOE will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan

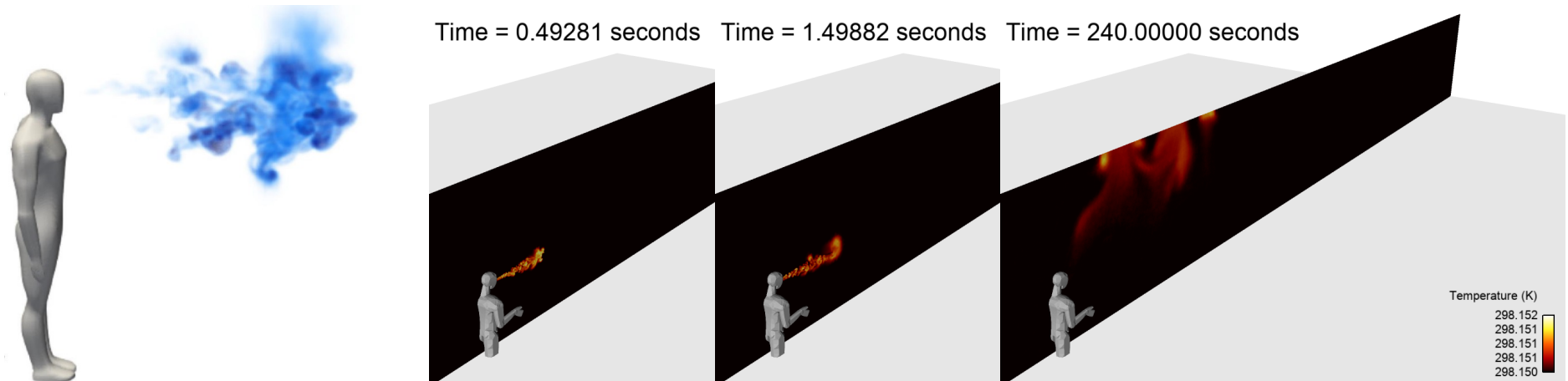
SAND2018-4536 PE





Simple Passive Scalar Transport

- The study of passive transport of nutrients or pollutants is important in biological and environmental applications (smoke, pathogens, etc.)



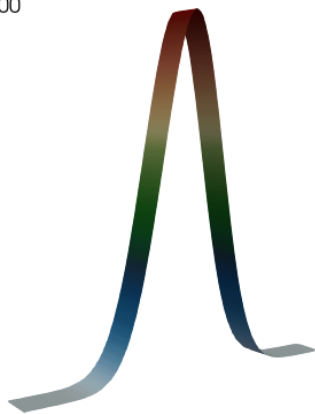
- Left, depicts a passive scalar representing a synthetic cough, Vuorinen et al. *"Modelling aerosol transport and virus exposure with numerical simulations in relation to SARS-CoV-2 transmission by inhalation indoors"*, Safety Science (2020)
- Recall that the conceptual model may require buoyancy effects, right, Domino, *"A case study on pathogen transport, deposition, evaporation and transmission: Linking high-fidelity computational fluid dynamics simulations to probability of infection"*, Int J. CFD (2021)



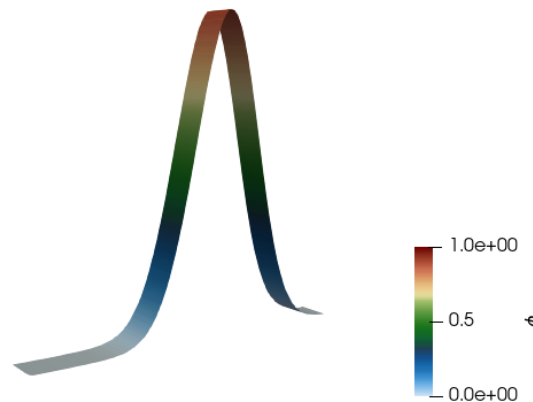
One Dimensional Temporal, Advection-Diffusion Equation Passive Scalar Transport

- Let us assume that the problem is defined in a one-dimensional configuration, and that the pollutant concentration is given by, $\phi(x,t)$
- Given an initial distribution of the pollutant $\phi(x,t = T_0)$, we can evaluate the concentration at any later time T_1 through solving a partial differential equation of the form:

Time: 0.000000



Time: 0.100000



$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} = \nu \frac{\partial^2 \phi}{\partial x^2}$$

Initial Condition:

- Gaussian

Boundary Condition:

- Domain is periodic

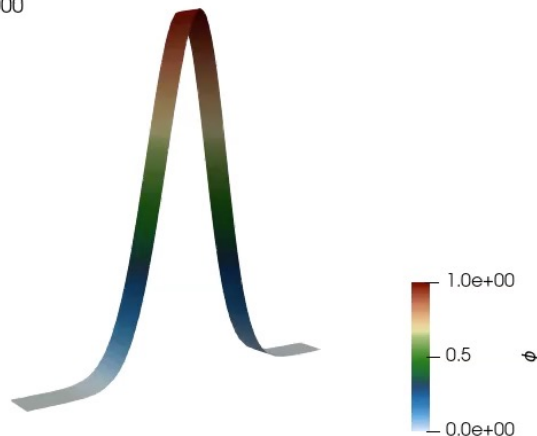
- Above, the air velocity v *is constant*
- Diffusion is characterized by a constant diffusivity ν , (m^2/s)



One Dimensional Temporal, Advection-Diffusion Equation Passive Scalar Transport

- Let us assume that the problem is defined in a one-dimensional configuration, and that the pollutant concentration is given by, $\phi(x,t)$
- Given an initial distribution of the pollutant $\phi(x,t = T_0)$, we can evaluate the concentration at any later time T_1 through solving a partial differential equation of the form:

Time: 0.000000



$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} = \nu \frac{\partial^2 \phi}{\partial x^2}$$

Initial Condition:

- Gaussian

Boundary Condition:

- Domain is periodic

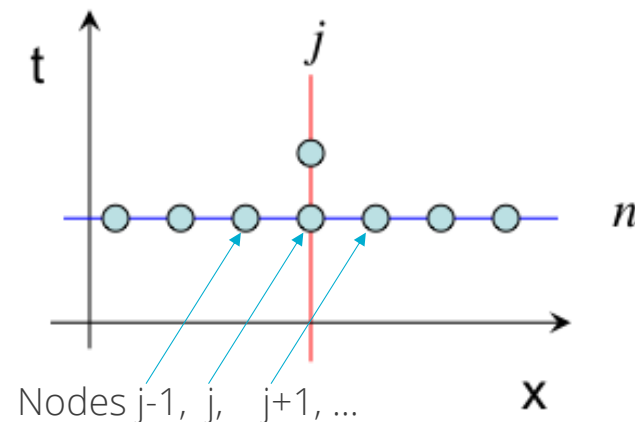
- Above, the air velocity v is constant
- Diffusion is characterized by a constant diffusivity ν , (m^2/s)



The Discretization Step

- The region of interest ($x_L \leq x \leq x_R$) is transformed into a discrete tessellation, i.e. the grid defined by the nodes x_j or the intervals between two nodes (volume) $x_{j+1} - x_j = \Delta x$
- A similar tessellation is also used to represent time discretely in terms of level t_n or intervals $t_{n+1} - t_n = \Delta \tau$
- The solution of the original advection-diffusion equation is only sought at the specific locations (x_j) and time (t_n)

$$\phi(x_j, t^n) = \phi_j^n$$





Approximations to the Continuous Derivative

- The governing equation must also be transformed into an equivalent discrete representation
- The simplest approach involves replacing each (continuous) derivative into a discrete relationship between the function ϕ represented at the grid nodes
- Recall the formal definition of derivative:

$$\frac{\partial \phi(x)}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{\phi(x + \Delta x) - \phi(x)}{\Delta x}$$

- Evaluated for a finite value of Δx

$$\frac{\partial \phi(x)}{\partial x} \approx \frac{\phi(x + \Delta x) - \phi(x)}{\Delta x} \quad (+\epsilon)$$

- Note: The approximate sign represent the effect of the discretization error consequence of a non-zero Δx



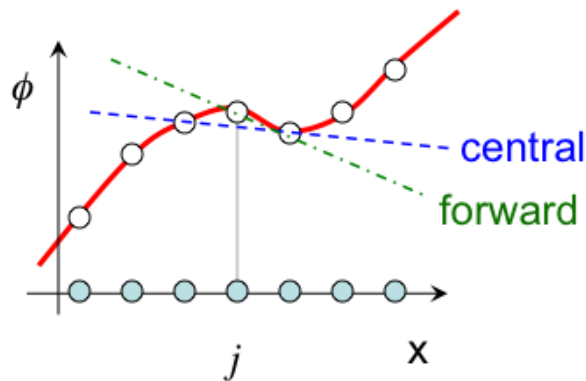
Geometric Interpretation

The formula introduced before,

$$\frac{\partial \phi(x)}{\partial x} \approx \frac{\phi(x + \Delta x) - \phi(x)}{\Delta x}$$

is only one of the possible representations of the derivative

It can be interpreted geometrically as the approximate slope of a smooth interpolant of the discrete values ϕ_j



$$\left(\frac{\partial \phi(x)}{\partial x} \right)_j \approx \frac{\phi_{j+1} - \phi_j}{x_{j+1} - x_j}$$

forward scheme

$$\left(\frac{\partial \phi(x)}{\partial x} \right)_j \approx \frac{\phi_{j+1} - \phi_{j-1}}{x_{j+1} - x_{j-1}}$$

central scheme



Algebraic System of Equations: Explicit

- Starting from the (continuous) mathematical model:

$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} = \nu \frac{\partial^2 \phi}{\partial x^2}$$

- The discretization step results in (discrete) computational model, for example using *forward-in-time* and *central-in-space* derivatives: FT-CS

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + v \frac{\phi_{j+1}^n - \phi_{j-1}^n}{2\Delta x} = \nu \frac{\phi_{j+1}^n - 2\phi_j^n + \phi_{j-1}^n}{\Delta x^2}$$

In this case the solution step is straightforward: $\phi_j^{n+1} = F(\phi_j^n, \phi_{j+1}^n, \phi_{j-1}^n, v, \nu, \Delta t, \Delta x)$

The scheme is **explicit and represents an example of a finite difference technique**: **Note: this scheme, for $\nu = 0$, is NOT STABLE (more later)**



Algebraic System of Equations: Implicit

Different choices of discretization techniques lead to more complex solution steps

- For example, *backward-in-time* and *central-in-space* derivatives: BT-CS

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + v \frac{\phi_{j+1}^{n+1} - \phi_{j-1}^{n+1}}{2\Delta x} = \nu \frac{\phi_{j+1}^{n+1} - 2\phi_j^{n+1} + \phi_{j-1}^{n+1}}{\Delta x^2}$$

- Here, the solution step is formally: $\phi_j^{n+1} = F(\phi_j^{n+1}, \phi_{j+1}^{n+1}, \phi_{j-1}^{n+1}, v, \nu, \Delta t, \Delta x)$
- Or, better written as a Matrix system: $A\phi^{n+1} = b$

This scheme is **implicit, i.e.**, requires a matrix inversion, **and represents an example of a finite difference technique**



Many Choices on Finite Difference Operators

Derivative	Accuracy	-5	-4	-3	-2	-1	0	1	2	3	4	5
1	2					-1/2	0	1/2				
	4				1/12	-2/3	0	2/3	-1/12			
	6			-1/60	3/20	-3/4	0	3/4	-3/20	1/60		
	8		1/280	-4/105	1/5	-4/5	0	4/5	-1/5	4/105	-1/280	
2	2					1	-2	1				
	4				-1/12	4/3	-5/2	4/3	-1/12			
	6			1/90	-3/20	3/2	-49/18	3/2	-3/20	1/90		
	8		-1/560	8/315	-1/5	8/5	-205/72	8/5	-1/5	8/315	-1/560	

Central finite difference

Derivative	Accuracy	0	1	2	3	4	5	6	7	8
1	1	-1	1							
	2	-3/2	2	-1/2						
	3	-11/6	3	-3/2	1/3					
	4	-25/12	4	-3	4/3	-1/4				
	5	-137/60	5	-5	10/3	-5/4	1/5			
	6	-49/20	6	-15/2	20/3	-15/4	6/5	-1/6		

Forward finite difference

Derivative	Accuracy	-8	-7	-6	-5	-4	-3	-2	-1	0
1	1								-1	1
	2							1/2	-2	3/2
	3						-1/3	3/2	-3	11/6

Backward finite difference

https://en.wikipedia.org/wiki/Finite_difference_coefficient



Properties of Numerical Methods

Outline:

- Consistency
- Stability, Von Neumann Analysis
- Convergence - Lax Equivalence Theorem
- Understanding Errors



High Level Concepts

A discretized mathematical model – a numerical model – is an approximation to the original, postulated, continuous model

- The properties of such numerical model have a strong effect on the computed solution

As a first step we will study three critical properties

- Consistency
- Accuracy
- Stability



Consistency

Definition: A discretization scheme for a mathematical model is consistent if it asymptotes to the original continuous form as the grid and timestep size tend to zero

- *Consistency is a property of the discretization, not of the numerical solution to the given problem*

Q: How to assess if a discretization technique is consistent?

A: Study the truncation error and verify that it vanishes as the grid and timestep size tend to zero.



Stability

Definition: A discretization scheme for a mathematical model is stable if numerical errors (e.g. round-off due to the precision in the computer representation) are not allowed to grow unbounded

- *Stability is a property of the discretization, not of the numerical solution to the given problem*

Q: How to assess if a discretization technique is stable?

A: This is fairly easy for linear PDEs with constant coefficients and no boundary conditions (periodic domains)



Stability Analysis

The simple discretization scheme (FT-CS) for the pure-advection problem is *consistent* but NOT *stable*

$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} = \cancel{\nu \frac{\partial^2 \phi}{\partial x^2}}_0$$

Recall it's explicit form:

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + v \frac{\phi_{j+1}^n - \phi_{j-1}^n}{2\Delta x} = 0$$

Can we understand why and, perhaps, design a different scheme that is stable?



Consistency + Stability Analysis

We can consider the effect of the truncation error as continuous terms in a modified differential equation

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + v \frac{\phi_{j+1}^n - \phi_{j-1}^n}{2\Delta x} = 0$$

Applying again a Taylor series expansion and plugging in the expression for the truncation error just found we obtain:

$$\begin{aligned} \frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} &= -\frac{\Delta t}{2} v^2 \frac{\partial^2 \phi}{\partial x^2} + O[(\Delta x)^2, (\Delta t)^2] \\ &= -\tilde{\nu} \frac{\partial^2 \phi}{\partial x^2} + O[(\Delta x)^2, (\Delta t)^2] \end{aligned}$$

- For a given Δx and Δt , note that we are not solving the pure advection equation, although the scheme is consistent (asymptotically)
- Physical viscosity (positive) has the effect of smoothing the solution; but here, the value is negative, and, therefore, **destabilizing**



Convergence

Definition: The numerical solution of a computational model approaches the exact if it asymptotes to the exact solution of the original continuous problem as the grid and timestep tend to zero

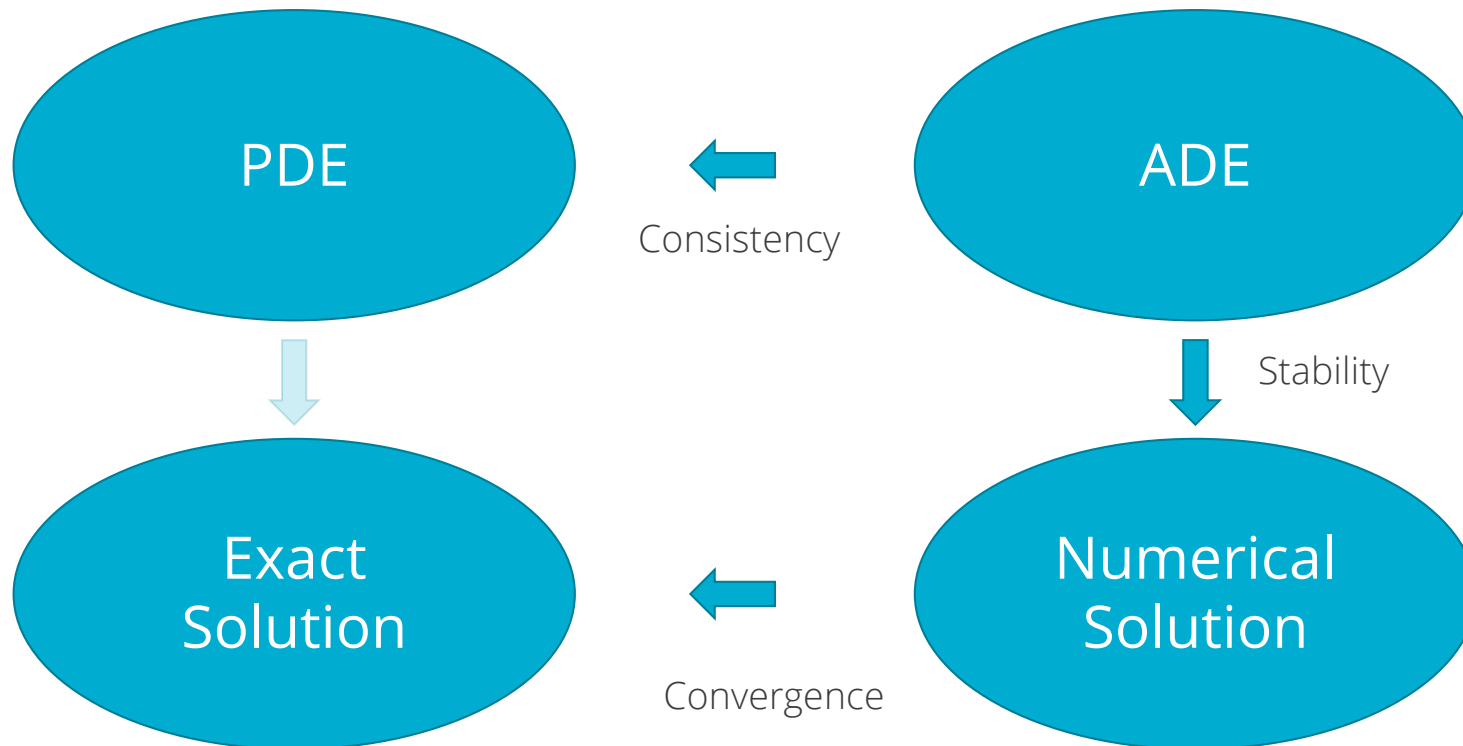
- *Convergence is a property of the solution*

Q: How to assess if the solution is converging?

A: This is formally possible only if the exact solution to the original problem is known!



Numerical Simulation Conundrum



- Convergence is required for a consistent and stable numerical method to recover the exact solution!



Lax Equivalence Theorem

Definition: For a well-posed linear problem discretized using a consistent method, stability is the necessary and sufficient condition for convergence

- In the case of a linear PDE, studying the properties of the computational method is sufficient to know that the exact solution can be computed; even before obtaining any numerical solution!
- For non-linear problems, the analysis is much harder and only weaker statements can be made
- Convergence is studied than directly and in addition to consistency and stability.



Origin of Numerical Errors

Consider again, the pure advection problem:

$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} = \cancel{\nu \frac{\partial^2 \phi}{\partial x^2}}_0$$

Recall, the *forward-in-time* and *central-in-space* scheme is consistent but unstable!

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + v \frac{\phi_{j+1}^n - \phi_{j-1}^n}{2\Delta x} = 0$$

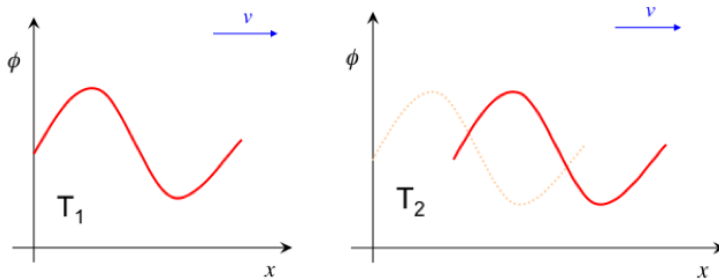
This PDE problem supports an exact solution with a simple physical interpretation

Let's use this finding to gain insight into the behavior of the numerical scheme

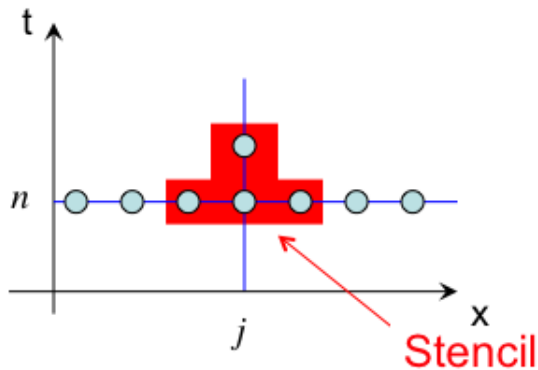


Evaluating the Exact Advection Equation Solution

Physical Interpretation: Information travels in the v-direction (positive)



Numerical Method: However, information is gathered from both j-1 and j+1 states



$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + v \frac{\phi_{j+1}^n - \phi_{j-1}^n}{2\Delta x} = 0$$



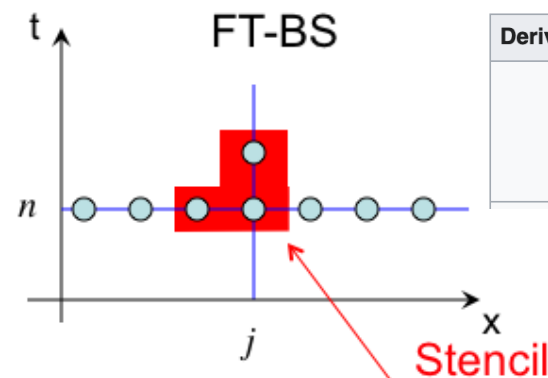
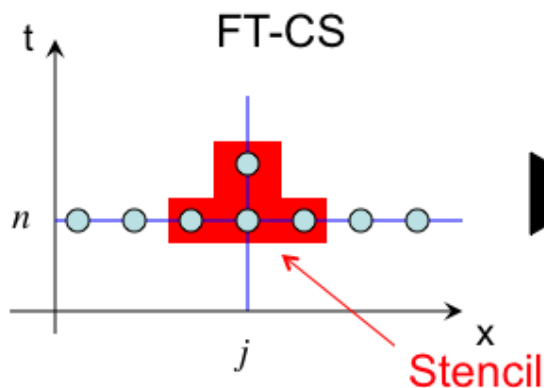
Exploiting the Directionality of the Flow

An alternative choice of spatial discretization provides a more physically-consistent discrete model: for example, if flow is moving from left to right, let's remove the $j+1$ contribution

- Consider a *forward-in-time* and *backward-in-space* differentiation (FT-BS)

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + v \frac{\phi_j^n - \phi_{j-1}^n}{\Delta x} = 0$$

Recall backward finite difference stencil



Derivative	Accuracy	-8	-7	-6	-5	-4	-3	-2	-1	0
1	1								-1	1
	2							1/2	-2	3/2
	3						-1/3	3/2	-3	11/6

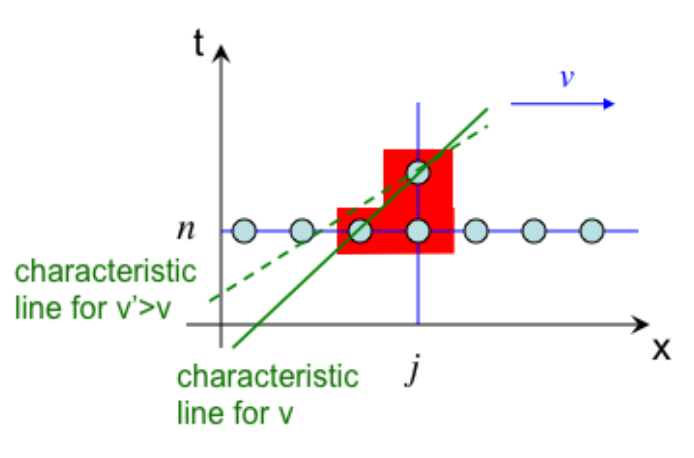
- This scheme is consistent and conditionally stable (see course reader) using Von Neumann stability analysis that shows $|\text{amplification factor}|$ is less than unity when $\frac{v\Delta t}{\Delta x} \leq 1$



Properties of the FT-BS

The scheme is consistent and conditionally stable. Why?

The characteristics lines (for fixed Δx and Δt) provide the physical picture:



Deriving the modified equation leads to a clear understanding of an underlying stabilization mechanism....



Modified Equation for the Forward-in-time, Backward-in-space

The FT-BS scheme is equivalent to:

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + \frac{v}{2\Delta x} (\phi_{j+1}^n - \phi_{j-1}^n) = \frac{v\Delta x}{2} \left(\frac{\phi_{j+1}^n - 2\phi_j^n + \phi_{j-1}^n}{\Delta x^2} \right)$$

and results in the modified equation:

$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} = \frac{v\Delta x}{2} \frac{\partial^2 \phi}{\partial x^2}$$

- The backward-in-space discretization leads to a positive dissipation, e.g. a stabilization of the scheme
- However, what happens if *velocity* < 0?



Introducing: An Upwind-based Scheme

Is it possible to simply switch between FT-BS ($v > 0$) and FT-FS ($v < 0$), while ensuring that the scheme is *consistent* and *conditionally* stable?

First, let's define an upwind and downwind operators:

$$v^+ = \frac{v + |v|}{2} \quad v^- = \frac{v - |v|}{2}$$

Furthermore, let's provide a general upwind scheme system as:

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + v^+ \frac{\phi_j^n - \phi_{j-1}^n}{\Delta x} + v^- \frac{\phi_{j+1}^n - \phi_j^n}{\Delta x} = 0$$

That can be reorganized as:

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + \frac{v}{2\Delta x} (\phi_{j+1}^n - \phi_{j-1}^n) = \frac{|v|\Delta x}{2} \left(\frac{\phi_{j+1}^n - 2\phi_j^n + \phi_{j-1}^n}{\Delta x^2} \right)$$



Modified Equation for the Upwind-based Scheme

Recall,

$$\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + \frac{v}{2\Delta x} (\phi_{j+1}^n - \phi_{j-1}^n) = \frac{|v|\Delta x}{2} \left(\frac{\phi_{j+1}^n - 2\phi_j^n + \phi_{j-1}^n}{\Delta x^2} \right)$$

and results in the modified equation:

$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} = \frac{|v|\Delta x}{2} \frac{\partial^2 \phi}{\partial x^2}$$

The above form provides stabilization that is positive for all real values of velocity

Stability can also be studied using Von Neumann analysis and results in the constraint that:

$$\frac{|v|\Delta t}{\Delta x} \leq 1$$



General Comments

- The consistency analysis and the derivation of the modified equation clearly illustrate the impact of numerical errors on the problem being solved
- The derivation of the upwind scheme shows that dissipation-like errors lead to stabilization
- What other errors can we expect to see using different discretization schemes?
- Let's re-visit the FT-CS advection system:

$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} = \nu \frac{\partial^2 \phi}{\partial x^2} \longrightarrow \frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + v \frac{\phi_{j+1}^n - \phi_{j-1}^n}{2\Delta x} = 0$$

- The truncation error for the FT-CS scheme is:

$$\epsilon = -\frac{\Delta t}{2} v^2 \left(\frac{\partial^2 \phi}{\partial x^2} \right)_j^n - v \frac{\Delta x^2}{6} \left(\frac{\partial^3 \phi}{\partial x^3} \right)_j^n + O \left[(\Delta x)^2, (\Delta t)^2 \right]$$

- Let's examine each separately



Model 1 and Model 2 Modified Equations

Model 1:
$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} - \alpha \frac{\partial^2 \phi}{\partial x^2} = 0$$

Model 2:
$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} + \beta \frac{\partial^3 \phi}{\partial x^3} = 0$$

Objective: Find the exact solution to these model equations in a simple periodic domain



Model 1

Postulate an expression for the solution: $\phi(x, t) = e^{pt} e^{ikx}$

Plug in the PDE: $\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} - \alpha \frac{\partial^2 \phi}{\partial x^2} = 0$

to obtain $pe^{pt} e^{ikx} + ivke^{pt} e^{ikx} + \alpha k^2 e^{pt} e^{ikx} = 0$

This is satisfied if $p = -ivk - \alpha k^2$, leading to the solution:

$$\phi(x, t) = e^{(-ivk - \alpha k^2)t} e^{ikx} = e^{ik(x-vt)} e^{-\alpha k^2 t}$$



Model 1

Postulate an expression for the solution: $\phi(x, t) = e^{pt} e^{ikx}$

Plug in the PDE: $\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} - \alpha \frac{\partial^2 \phi}{\partial x^2} = 0$

to obtain $pe^{pt} e^{ikx} + ivke^{pt} e^{ikx} + \alpha k^2 e^{pt} e^{ikx} = 0$

This is satisfied if $p = -ivk - \alpha k^2$, leading to the solution:

$$\phi(x, t) = e^{(-ivk - \alpha k^2)t} e^{ikx} = e^{ik(x-vt)} e^{-\alpha k^2 t}$$



Model 2

Postulate an expression for the solution: $\phi(x, t) = e^{pt} e^{ikx}$

Plug in the PDE: $\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} + \beta \frac{\partial^3 \phi}{\partial x^3} = 0$

to obtain $pe^{pt} e^{ikx} + ivke^{pt} e^{ikx} - \beta ik^3 e^{pt} e^{ikx} = 0$

This is satisfied if $p = -ivk + \beta ik^3$, leading to the solution:

$$\phi(x, t) = e^{(-ivk + \beta ik^3)t} e^{ikx} = e^{ik[x - (v - \beta k^2)t]} = e^{ik(x - wt)}$$



Model 2

Postulate an expression for the solution: $\phi(x, t) = e^{pt} e^{ikx}$

Plug in the PDE: $\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} + \beta \frac{\partial^3 \phi}{\partial x^3} = 0$

to obtain $pe^{pt} e^{ikx} + ivke^{pt} e^{ikx} - \beta ik^3 e^{pt} e^{ikx} = 0$

This is satisfied if $p = -ivk + \beta ik^3$, leading to the solution:

$$\phi(x, t) = e^{(-ivk + \beta ik^3)t} e^{ikx} = e^{ik[x - (v - \beta k^2)t]} = e^{ik(x - wt)}$$



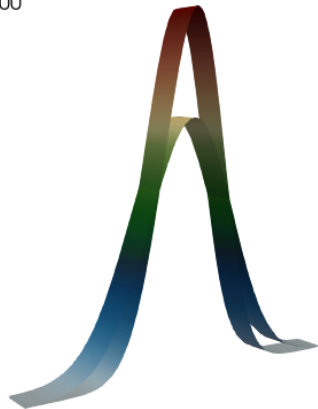
Model 1: Dissipative-like Error

Review:

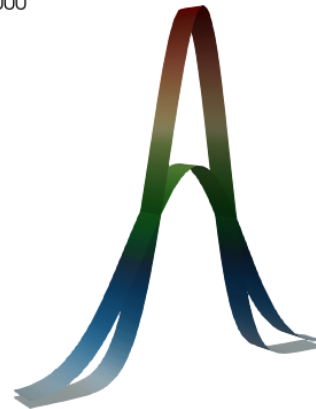
$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} - \alpha \frac{\partial^2 \phi}{\partial x^2} = 0$$

$$\phi(x, t) = e^{(-ivk - \alpha k^2)t} e^{ikx} = e^{ik(x-vt)} e^{-\alpha k^2 t}$$

Time: 1.000000



Time: 2.000000





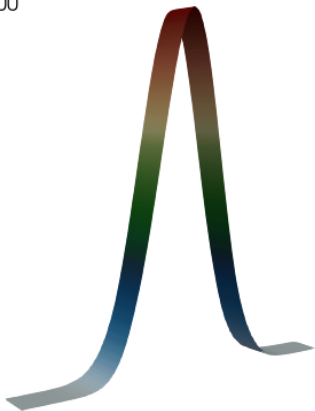
Model 2: Dispersion-like Error

Review:

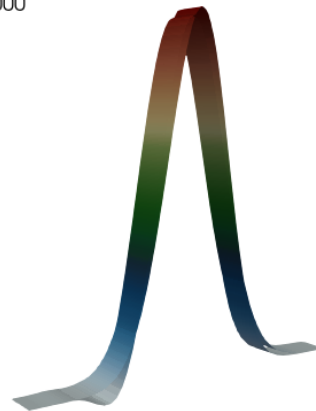
$$\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} + \beta \frac{\partial^3 \phi}{\partial x^3} = 0$$

$$\phi(x, t) = e^{(-ivk + \beta ik^3)t} e^{ikx} = e^{ik[x - (v - \beta k^2)t]} = e^{ik(x - wt)}$$

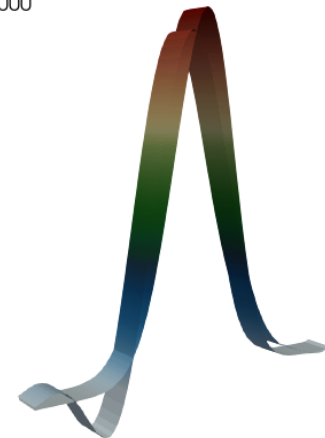
Time: 1.000000



Time: 4.000000



Time: 8.000000



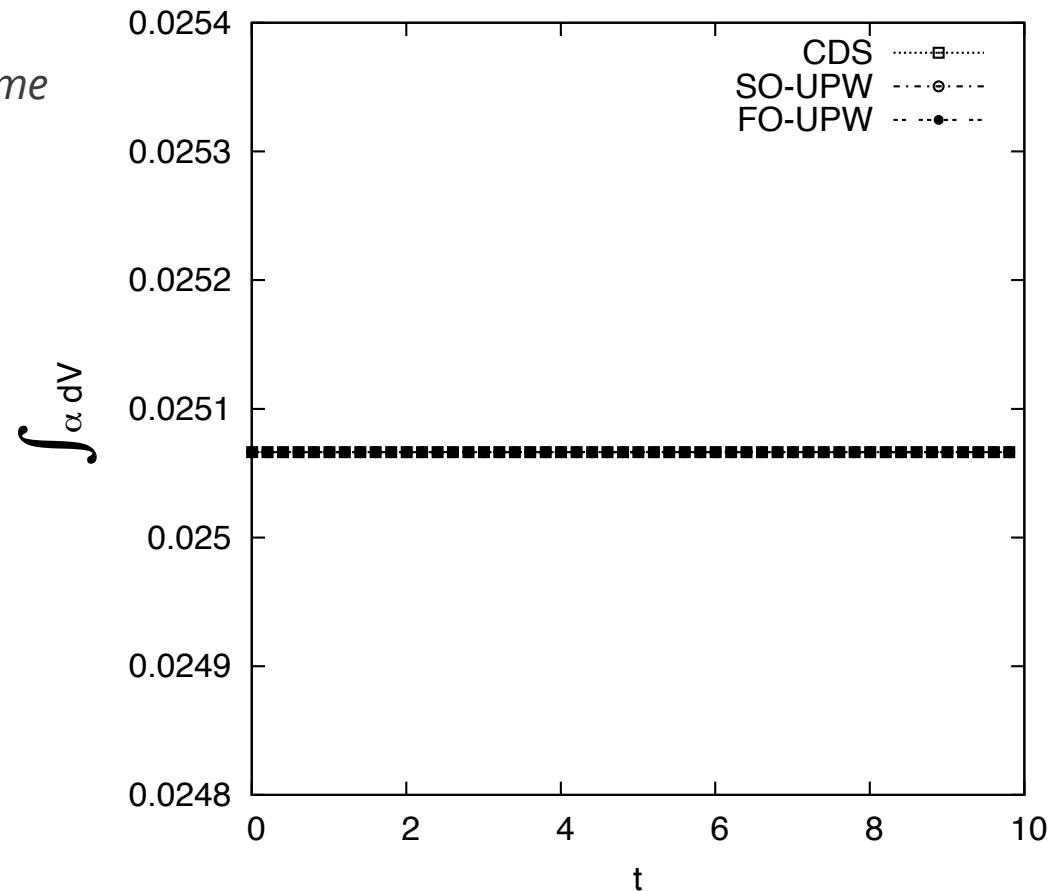


A Note on Conservation: ϕ

For our fully implicit *backward-in-time* and *central-in-space* Nalu finite volume approach:

Integration ϕ over the full domain, for any advection operator choice, results in:

- Perfect conservation





Derivation of a Energy Quantity

Starting with our model equation: $\frac{\partial \phi}{\partial t} + v \frac{\partial \phi}{\partial x} = \cancel{\nu \frac{\partial^2 \phi}{\partial x^2}}$ $\downarrow$
0

And scale by ϕ : $\phi \left(\frac{\partial \phi}{\partial t} + u_j \frac{\partial \phi}{\partial x_j} \right) = 0$

Results in the modified equation, $\frac{\partial \frac{\phi^2}{2}}{\partial t} + u_j \frac{\partial \frac{\phi^2}{2}}{\partial x_j} = 0 \longrightarrow \frac{\phi^2}{2}$

that mimics an energy, $\frac{u_i u_i}{2}$

Let's see how this quantity is conserved....

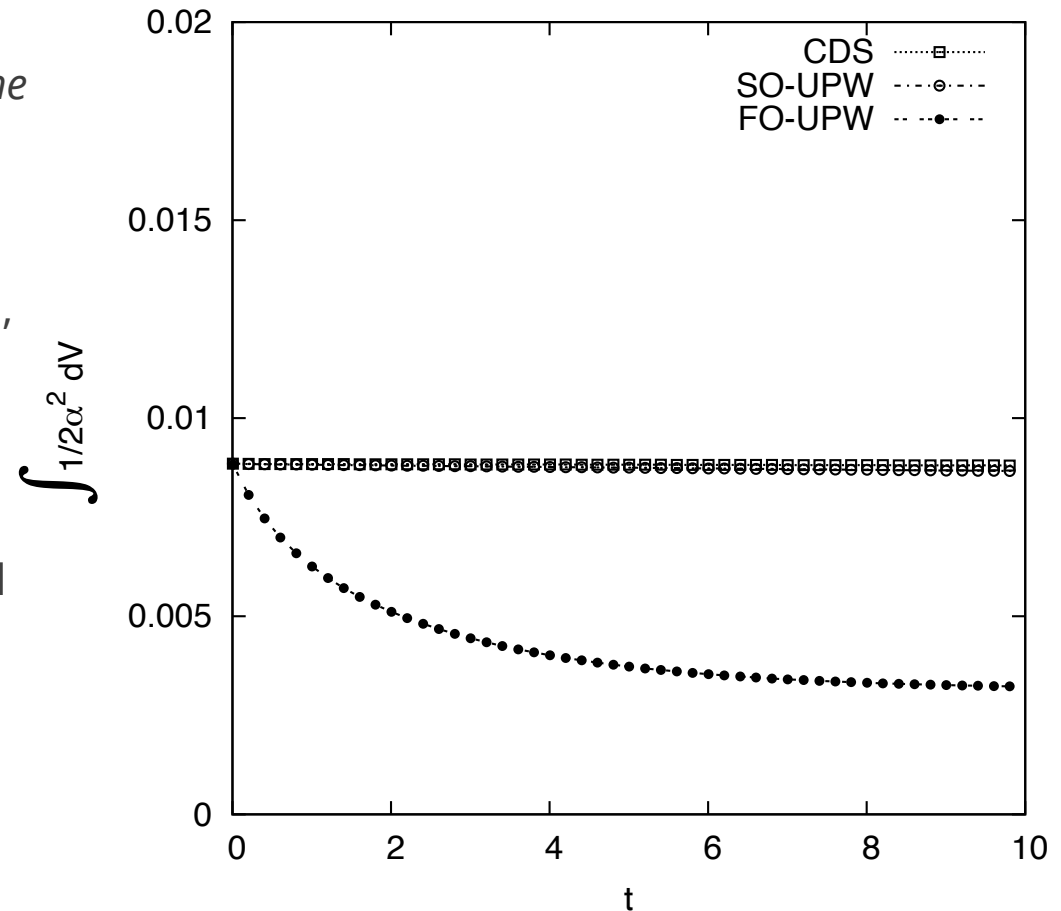


A Note on Conservation: $\phi^2/2$

For our fully implicit *backward-in-time* and *central-in-space* Nalu finite volume approach:

Integration $\phi^2/2$ over the full domain, for any advection operator choice, results in:

- Lack of conservation for the upwinded approaches
- Ideal conservation for the central operator
- Energy-method where the PDE is derived for the energy-variable, see course reader





Final Notes

- Studying the numerical properties of the discretization technique is critical to set the proper expectations on the solutions
- Classical tools such as Von Neumann stability analysis are easy to use but not generalizable
- Other properties (beyond consistency, stability, etc.) can be critical.
 - We will introduce *conservation*, *boundedness* and *symmetry*.
- Good understanding of the physics of the problem leads to better choices for the numerical schemes